

PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Itgalpura, Rajankunte, Yelahanka, Bengaluru - 560064

AI GPU Summer Internship Program

CSS7000 INTERNSHIP REPORT

Submitted by

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Under the guidance of,

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BACHELOR OF TECHNOLOGY

IN

**COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence and Machine Learning)**



**PRESIDENCY UNIVERSITY
BENGALURU
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PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING DEPARTMENT OF AI & ROBOTICS

BONAFIDE CERTIFICATE

Certified that this report “**AI GPU summer Internship program**” is a bonafide work of **AAKASH . R (20241CAI0039)**, who has successfully carried out the internship work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in PRESIDENCY SCHOOL OF ARTIFICIAL INTELLIGENCE & ADVANCED COMPUTING , **AI & Robotics** during 2026-2027.

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DECLARATION

I am a student of Pre - Final year B.Tech in COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence and Machine Learning), at Presidency University, Bengaluru, named, **AAKASH . R** hereby declare that the internship work titled “**AI GPU summer Internship program**” has been independently carried out by me and submitted in partial fulfillment for the award of the degree of B.Tech in COMPUTER SCIENCE AND ENGINEERING during the academic year of 2026-2027. Further, the matter embodied in the internship has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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INTERNSHIP COMPLETION CERTIFICATE

NVIDIA Accelerated AI Centre of Excellence

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This certifies that

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has successfully completed the AI GPU Summer Internship Program

August 2026

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Aakash. R

ABSTRACT

The AI GPU Summer Internship – 2026 provided an intensive practical learning experience in artificial intelligence, machine learning, deep learning, GPU computing, and modern AI workflows. Conducted through the AI Centre of Excellence (Accelerated by NVIDIA) at Presidency University, the ten-instructional-day program combined theory, guided coding, GPU laboratory sessions, mentored project work, and final assessment. The training included machine learning fundamentals, GPU-accelerated workflows, PyTorch, convolutional neural networks, GPU profiling, transformer architectures, parameter-efficient fine-tuning, generative AI, and GPU optimization concepts.

The major project developed as part of the internship was the Smart Crop Recommendation System. The system addresses the problem of selecting a suitable crop based on soil and environmental conditions. It uses seven input features—Nitrogen, Phosphorus, Potassium, temperature, humidity, soil pH, and rainfall—to predict one of 22 crop categories. The project uses a Crop Recommendation Dataset obtained from Kaggle containing 2,200 instances. The data was explored, preprocessed, label encoded, and divided into training and testing sets using an 80:20 split.

Five supervised learning algorithms—Decision Tree, Random Forest, K-Nearest Neighbors, Support Vector Machine, and Gaussian Naive Bayes—were implemented and compared. Random Forest achieved the best performance among the evaluated models and was selected for deployment. The trained model was saved using Joblib and integrated into a Streamlit web application that accepts agricultural parameters and provides a crop recommendation, prediction confidence, and crop-related information. The project demonstrates how machine learning can support data-driven agricultural decision-making and provides a practical example of applying the concepts learned during the AI GPU internship.

Keywords: Artificial Intelligence, Machine Learning, GPU Computing, NVIDIA, PyTorch, Crop Recommendation, Random Forest, Streamlit, Precision Agriculture.

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Chapter 1: Introduction to the Internship

The rapid growth of artificial intelligence has increased the importance of high-performance computing platforms capable of handling large datasets, deep neural networks, and computationally intensive workloads. The AI GPU Summer Internship – 2026 was designed to provide practical exposure to these technologies through a structured combination of lectures, coding workshops, GPU laboratories, mentored project work, and assessment.

According to the internship offer, the program was organized by the Presidency School of Artificial Intelligence and Advanced Computing in direct collaboration with the NVIDIA Accelerated AI Centre of Excellence. The assigned cohort for this report was Slot 3, scheduled from 13 July to 25 July 2026, with ten instructional days. The assigned daily session was 09:00 AM to 12:00 PM in the D-Block GPU Computing Lab cluster.

The internship therefore served not only as a short-term training program but also as a bridge between classroom concepts and practical AI engineering. The project work provided an opportunity to apply machine learning methods to a real-world problem while following a complete data-to-deployment workflow.

Parameter	Internship Detail
Program	AI GPU Summer Internship – 2026
Centre	AI Centre of Excellence (Accelerated by NVIDIA)
Cohort	Slot 3
Dates	13 July – 25 July 2026
Duration	10 instructional days (3 hours/day, 30 contact hours)
Academic credit	2 transferable credits
Project	Smart Crop Recommendation System

Chapter 2: Internship Overview and Training Structure

The internship was structured as a 30-hour intensive workflow. The official program structure allocated 20% to lecture/theory, 20% to guided coding workshops, 33% to GPU hands-on laboratories, 20% to mentored project work, and 7% to final assessment and presentations. This balance placed greater emphasis on hands-on learning and project execution.

Component	Allocation	Approx. Hours
Lecture / Theory	20%	6 hours
Guided Coding Workshops	20%	6 hours
GPU Hands-on Labs	33%	10 hours
Mentored Project Work	20%	6 hours
Final Assessment / Presentation	7%	2 hours
Total	100%	30 hours

Assessment Structure

The assessment strategy was continuous, checking daily practical skills as well as final Capstone project delivery. The daily submissions validated the code written during lab sessions, while the class quiz tested conceptual knowledge. The major project presentation allowed students to showcase their crop recommendation applications.

Assessment Component	Weight
Daily GPU lab submissions	35%
Week 1 class test / quiz	10%
Capstone project	50%
Peer review and participation	5%

Chapter 3: NVIDIA-Oriented GPU Training

A major aspect of the internship was exposure to GPU computing for artificial intelligence workloads. The official training documentation states that students were provided access to a dedicated container instance on an NVIDIA H200 Tensor Core GPU cluster through JupyterHub. The H200 is described in the program material as being based on the Hopper GH100 architecture, with 141 GB of HBM3e memory, 4.8 TB/s memory bandwidth, 3,958 FP8 Tensor TFLOPS, and integrated Transformer Engines.

Item	Description
GPU platform	NVIDIA H200 Tensor Core GPU
Architecture	Hopper (GH100)
Interface	JupyterHub / dedicated container
Memory	141 GB HBM3e
Memory bandwidth	4.8 TB/s
AI focus	Deep learning, transformers, LLMs, generative AI and optimization

Why GPUs Matter for AI

- GPUs provide highly parallel computation suitable for matrix and tensor operations used in machine learning.
- Deep learning models contain large numbers of parameters and operations that can benefit from parallel hardware.
- GPU memory bandwidth is important when training and running data-intensive models.
- Modern AI workloads such as transformers and generative models can require substantially more compute than conventional CPU-only workflows.
- GPU-oriented tools also help developers profile, optimize, and understand the performance of AI programs.

Chapter 4: Training Modules and Technical Learning

The ten instructional days were divided into two broad phases. Week 1 focused on AI foundations and deep learning essentials, while Week 2 moved toward advanced AI, large language models, generative AI, optimization, and capstone execution.

Day	Module / Focus Description
Day 1	Orientation and environment setup on JupyterHub.
Day 2	Machine Learning fundamentals and accelerated Scikit-learn on GPU.
Day 3	Neural Networks from Scratch and PyTorch basics on NVIDIA H200.
Day 4	Convolutional Neural Networks (CNNs) and GPU profiling with Nsight.
Day 5	Hands-on Google Antigravity and foundation quiz assessment.

Selected Technical Themes

- PyTorch was introduced as a framework for building and training neural networks.
- CNN concepts connected deep learning with image-based applications.
- Nsight profiling introduced performance analysis of GPU workloads.

Chapter 4: Training Modules and Technical Learning

The second week of the training program expanded on modern large-scale architectures and deployment methodologies. The modules covered the mathematical frameworks of transformers, parameters optimization, and generative pipelines.

Day	Module / Focus Description
Day 6	Transformer architecture deep dive: attention mechanisms and tokenization.
Day 7	LLM fine-tuning with LoRA and PEFT using Hugging Face on H200.
Day 8	Generative AI using Stable Diffusion XL pipelines.
Day 9	GPU optimization (mixed precision, FlashAttention) and capstone execution.
Day 10	Capstone presentation, peer review and closing ceremony.

From Classical ML to Deep Learning

The Capstone project itself primarily uses classical supervised machine learning rather than a deep neural network. This is an important engineering decision: the appropriate model depends on the dataset, problem structure, computational requirements, and expected deployment environment. The internship therefore helped frame model selection as a problem-solving decision rather than an assumption that the newest or largest model is always best.

Chapter 5: Project Overview

The Smart Crop Recommendation System is a machine learning-based decision-support application designed to recommend a suitable crop from agricultural input conditions. The system analyzes soil nutrients and environmental conditions and predicts one of the crop categories represented in the dataset.

Feature	Role in the System
Nitrogen (N)	Amount of nitrogen present in soil (essential for leaf growth)
Phosphorus (P)	Amount of phosphorus present in soil (essential for root development)
Potassium (K)	Amount of potassium present in soil (essential for crop disease resistance)
Temperature	Average environmental temperature in °C
Humidity	Relative humidity of the air in %
Soil pH	Acidity / alkalinity level of the soil
Rainfall	Annual rainfall level in mm

Project Significance

Agricultural decisions depend on multiple interacting factors. A machine learning model can evaluate several variables together and provide a consistent data-driven recommendation. The project therefore demonstrates a practical application of supervised classification to precision agriculture, showing how sensor-based features translate into recommendations.

Chapter 6: Problem Statement and Objectives

6.1 Problem Statement

Agricultural productivity is strongly influenced by choosing crops compatible with soil properties and prevailing environmental conditions. Traditional crop selection may depend on experience, local knowledge, and seasonal assumptions. Changes in soil fertility, temperature, humidity, rainfall, and other conditions can make the decision more difficult. The project addresses the need for an intelligent system capable of analyzing multiple agricultural parameters simultaneously and recommending a suitable crop.

6.2 Objectives

- Analyze soil nutrient and environmental data from the Crop Recommendation Dataset.
- Perform data preprocessing and exploratory data analysis (EDA).
- Train multiple supervised classification algorithms (Decision Tree, Random Forest, KNN, SVM, Naive Bayes).
- Compare model performance and identify the best-performing algorithm.
- Deploy the selected model through a user-friendly Streamlit web application.
- Provide prediction confidence and crop-related information to support informed decision-making.

6.3 Scope

The present scope covers crop recommendation using the seven input parameters available in the selected dataset. It includes data processing, model development, comparative evaluation, and a basic web deployment. The current system does not include live weather retrieval, fertilizer scheduling, irrigation planning, disease diagnosis, or field-level sensor integration; these are treated as possible future extensions.

Chapter 7: Dataset Description

The project uses the Crop Recommendation Dataset obtained from Kaggle. The dataset contains 2,200 instances and eight attributes: seven input features and one target variable. The target variable represents the recommended crop and contains 22 crop classes.

Attribute	Description	Type
N	Amount of nitrogen present in soil	Numeric (Integer)
P	Amount of phosphorus present in soil	Numeric (Integer)
K	Amount of potassium present in soil	Numeric (Integer)
temperature	Average environmental temperature in °C	Numeric (Float)
humidity	Relative humidity of the air in %	Numeric (Float)
ph	Acidity or alkalinity level of the soil	Numeric (Float)
rainfall	Annual rainfall level in mm	Numeric (Float)
label	Recommended crop / target class (22 crop types)	Categorical (String)

Crop Classes

The dataset contains 22 crop categories, including Rice, Maize, Chickpea, Kidney Beans, Pigeon Peas, Moth Beans, Mung Bean, Black Gram, Lentil, Pomegranate, Banana, Mango, Grapes, Watermelon, Muskmelon, Apple, Orange, Papaya, Coconut, Cotton, Jute, and Coffee. The classes are balanced, with 100 instances for each crop type, making it a reliable dataset for classification.

Chapter 8: Methodology

The project follows a structured machine learning pipeline beginning with data collection and ending with deployment. The stages were designed to maintain a clear separation between raw data, preprocessing, model training, evaluation, and user-facing prediction.

Stage	Description
1. Data Collection	Obtain the Crop Recommendation Dataset from Kaggle.
2. Data Inspection	Inspect structure and statistics using Pandas head(), info(), and describe().
3. Quality Check	Check missing values and understand feature distributions.
4. Target Encoding	Convert categorical crop labels into numerical representations.
5. Train-Test Split	Use an 80:20 split for model training and unseen testing.
6. EDA	Analyze distributions and feature relationships using charts and correlation analysis.
7. Model Training	Train Decision Tree, Random Forest, KNN, SVM, and Gaussian Naive Bayes.
8. Evaluation	Compare accuracy and use confusion matrix and classification reports.
9. Model Selection	Select Random Forest based on the reported best performance.
10. Deployment	Save the model with Joblib and integrate it into Streamlit web application.

Chapter 9: Exploratory Data Analysis and Preprocessing

9.1 Data Preprocessing Details

Before feeding the data into machine learning algorithms, the raw dataset was preprocessed. A missing-value check was performed using Pandas `isnull().sum()`. The check confirmed that no missing values were present in any of the seven input columns, so no data imputation was necessary. Next, the target crop labels (e.g. 'rice', 'maize') were categorical, so a Label Encoder was used to map each of the 22 categories to a unique integer from 0 to 21.

9.2 Train-Test Split Configuration

To evaluate model generalization, the preprocessed dataset was split into training and testing sets. An 80:20 split ratio was applied using Scikit-learn's `train_test_split()`. The training subset (1,760 samples) was used for training the classifiers, while the testing subset (440 samples) was reserved for performance evaluation.

9.3 Feature Distribution Analysis

Exploratory analysis was conducted to study feature properties. Descriptive statistics were computed using the `describe()` method, revealing a wide range of values (e.g. nitrogen ranging from 0 to 140, and rainfall ranging from 20mm to 298mm). Correlation analysis was performed to inspect linear relationships, showing high correlation between Potassium (K) and Phosphorus (P), which is common in agricultural features.

Chapter 10: Machine Learning Algorithms

Five supervised learning algorithms were selected and trained to solve the crop recommendation task. The models represent tree-based, distance-based, margin-based, and probabilistic approaches.

1. Decision Tree Classifier: A classifier that creates recursive split rules based on feature purity (Gini or Entropy). Interpretability is high, but trees can overfit easily.

2. Random Forest Classifier: An ensemble classifier combining multiple Decision Trees. It uses bootstrap aggregation (bagging) and random feature splits to reduce variance and overfitting. *This model achieved the highest performance.*

3. K-Nearest Neighbors (KNN): A distance-based classifier that assigns a class according to the majority vote of the K closest data points.

4. Support Vector Machine (SVM): A margin-based classifier that finds the optimal separating hyperplane between classes in a high-dimensional kernel space.

5. Gaussian Naive Bayes: A probabilistic classifier that applies Bayes' theorem under the assumption of feature independence using Gaussian distributions.

Algorithm	Main Principle	Reported Performance
Decision Tree	Tree-based decision rules	Good
Random Forest	Ensemble of decision trees	Best Performance (Selected)
KNN	Nearest-neighbor voting	Good
SVM	Maximum-margin hyperplane	Moderate
Gaussian Naive Bayes	Probabilistic classification	Good

Chapter 11: Model Training and Evaluation

11.1 Model Training Procedures

Each classifier was trained on the training subset generated after preprocessing. The purpose of training was to learn relationships between the seven agricultural input features and the crop labels. Since the problem is multiclass classification, each model had to learn boundaries distinguishing among the 22 possible crop categories.

11.2 Evaluation Metrics & Selection

The models were evaluated using the test partition of the data. Accuracy was the primary metric used to compare the models. Additionally, confusion matrices and classification reports (precision, recall, F1-score) were generated to ensure correct predictions across all 22 classes without bias toward specific crops.

The project documentation reports that the **Random Forest Classifier** achieved the highest prediction accuracy among the five evaluated models. It was selected as the final classifier for deployment. The selection illustrates the importance of evaluating multiple candidate algorithms rather than assuming that one method will always perform best.

11.3 Overfitting Mitigation

To minimize overfitting, the project utilized a separate testing set for evaluation. The ensemble approach of the Random Forest model, which averages predictions across multiple randomized trees, also naturally increases robustness and improves generalization on unseen test instances compared to individual decision trees.

Chapter 12: System Architecture and Implementation

12.1 High-Level Architecture

The application follows a standard modular design. The user interface allows users to enter inputs, which are preprocessed, formatted, and passed to the trained model. The model generates a crop prediction, which is returned to the user with confidence scores.

Technology / Tool	Purpose
Python	Core programming language for data pipelines and modeling
Pandas / Numpy	Data loading, manipulation, and numerical processing
Scikit-learn	Machine learning algorithms training, scaling, and evaluation
Joblib	Model persistence (serializing and saving the trained model to disk)
Streamlit	Web application framework for user interface and deployment
Kaggle dataset	Agricultural training source data (2,200 samples)

12.2 Model Persistence Details

After training and hyperparameter selection, the Random Forest model was serialized and saved using ``joblib.dump()`. This persists the model parameters on disk, allowing the web server to load the trained model instantly using `joblib.load()`. Without wasting time retraining the model at startup.`

Chapter 13: Streamlit Deployment and User Workflow

The project was deployed as an interactive web application using Streamlit. Streamlit is a lightweight Python framework that allows data scientists to build UI screens without writing separate HTML, CSS, or JavaScript code.

User Workflow Step-by-Step

1. ****Access the Portal****: Open the application in a web browser.
2. ****Enter Soil Nutrients****: Input values for Nitrogen (N), Phosphorus (P), and Potassium (K) using sliders or boxes.
3. ****Enter Environmental Parameters****: Input local temperature, relative humidity, and annual rainfall levels.
4. ****Enter Soil Acidity****: Specify the pH of the soil.
5. ****Submit Data****: Click the 'Predict Recommended Crop' button.
6. ****View Results****: Read the recommended crop name, prediction confidence, and relevant agricultural tips.

Deployment Advantages

- ****User Accessibility****: Provides an interactive dashboard for non-programming users (like farmers or agronomists).
- ****Inference Efficiency****: Separates model training from prediction runtime, ensuring instantaneous recommendations.
- ****Flexible Testing****: Allows rapid user testing with arbitrary agricultural inputs to evaluate model behavior.
- ****HPC Context****: Serves as a practical demonstration of translating AI models trained on GPU-accelerated clusters to user endpoints.

Chapter 14: Results and Discussion

14.1 Comparative Analysis Summary

The Smart Crop Recommendation System was successfully trained and evaluated. The five classifiers—Decision Tree, Random Forest, KNN, SVM, and Gaussian Naive Bayes—were tested under identical conditions. The Random Forest Classifier consistently achieved the best performance across all crop classes.

14.2 Model Interpretation

The results demonstrate that the ensemble strategy used by Random Forest is highly effective for tabular agricultural data. By constructing a large collection of decision trees during training and outputting the consensus class vote, Random Forest can model complex, nonlinear feature interactions (such as the combined impact of high temperature and low rainfall on soil moisture) without overfitting. This makes it more robust than individual Decision Trees, which are highly sensitive to small data variations.

14.3 Practical Output & Deployment Validation

The Streamlit application successfully integrates the persistent Joblib model. In testing, the application responded to input parameter changes instantly, returning logical predictions (e.g. predicting 'Rice' under high water and nitrogen levels, and 'Chickpea' under dry, low-water conditions). This validates the utility of the system as an interactive agricultural decision-support tool.

Chapter 14: Results and Discussion

14.4 Performance Notes & Qualitative Analysis

A qualitative comparison of the models highlights their structural strengths and weaknesses when applied to this multiclass classification task. The distance-based KNN model performed well but suffered from latency scaling when the reference training size grew. The SVM model required scaling normalization and careful hyperparameter tuning to converge, while Gaussian Naive Bayes provided a fast baseline but was limited by its strong feature-independence assumption.

Model	Primary Strength	Inference Latency
Decision Tree	Highly interpretable logic pathways	Very Low (< 1ms)
Random Forest	Ensemble average prevents overfitting	Low (1-3ms)
K-Nearest Neighbors	Simple non-parametric approach	Medium (dependent on database size)
Support Vector Machine	Effective in high-dimensional spaces	Low-Medium (kernel math)
Gaussian Naive Bayes	Fast probabilistic calculations	Very Low (< 1ms)

Important Reporting Note: The project source report identifies Random Forest as the highest-performing model but does not provide a numeric accuracy value in the supplied document. Therefore, this report intentionally does not invent a percentage accuracy, adhering strictly to the verified qualitative outputs provided in your source materials.

Chapter 15: Limitations and Future Scope

15.1 Current Limitations

- **Feature Constraints**: The system relies on the seven features available in the dataset, omitting other parameters such as organic carbon, soil texture, seed quality, or solar radiation.
- **Manual Input**: The application requires manual data entry rather than automatically retrieving live environmental conditions.
- **Narrow Focus**: The system recommends crops but does not provide complete crop cycle suggestions, fertilizer schedules, or disease management advice.
- **Qualitative Bounds**: Quantitative model comparisons are limited by the absence of numeric accuracy values in the source documentation.

15.2 Future Enhancements

- **Live Weather API**: Integrate real-time weather coordinates to fetch temperature, humidity, and rainfall automatically based on location.
- **Fertilizer Recommender**: Add secondary models to suggest optimal N-P-K fertilizer ratios for the recommended crop.
- **Irrigation Scheduling**: Build timer-based irrigation recommendation modules based on relative soil moisture levels.
- **Mobile App and Localization**: Port the Streamlit code to mobile frameworks and add local language support for farmers in different regions.
- **IoT Integration**: Integrate physical soil sensors (e.g. NPK sensors, moisture probes) directly into the app for automated real-time prediction.

Chapter 16: Learning Outcomes

The internship provided extensive experience in modern data engineering and parallel systems. Key learning outcomes include: (1) Understanding supervised classification pipelines and multiclass modeling; (2) Preprocessing, encoding, and dividing datasets; (3) Training and comparing multiple models to select the optimal classifier; (4) Developing web interfaces using Streamlit; and (5) Gaining exposure to GPU acceleration paradigms (NVIDIA H200) and GPU profiling.

Chapter 17: Conclusion and References

The AI GPU Summer Internship – 2026 provided a structured foundation in parallel AI architectures and software stacks. The Capstone project—Smart Crop Recommendation System—demonstrated a complete ML workflow, using a Random Forest Classifier to recommend crop classes based on soil and environmental features. Future enhancements can extend the system towards real-time weather fetching, fertilizer suggestions, and mobile deployment.

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